PHYS 209: Aspects of Modern Physics

Module 4: Qubits

Lecture 4: Quantum Communication

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Unitary Operators

- Pauli Matrices:

$$\hat{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad \hat{X} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad \hat{Y} = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \qquad \hat{Z} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

  - These are examples of unitary operators.
- Unitary operators $\hat{U}$ are defined as

$$\hat{U}\hat{U}^{\dagger} = \hat{U}^{\dagger}\hat{U} = \hat{I}$$

  - Reminder:

$$\hat{U}^{\dagger} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{\dagger} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{T*} = \begin{bmatrix} a & c \\ b & d \end{bmatrix}^{*} = \begin{bmatrix} a^{*} & c^{*} \\ b^{*} & d^{*} \end{bmatrix}$$

- Time Evolution Operator $\hat{U} = e^{-i\hat{H}t/\hbar}$ is also unitary.

Why are Unitary Operators important?

$$\hat{U}|\psi\rangle = |\psi'\rangle, \qquad \hat{U}|\phi\rangle = |\phi'\rangle, \qquad \hat{U}^\dagger\hat{U} = \hat{I}$$

- Unitary operators preserve probabilities:

$$\langle\psi'|\psi'\rangle = \langle\psi|\hat{U}^\dagger\hat{U}|\psi\rangle = \langle\psi|\hat{I}|\psi\rangle = \langle\psi|\psi\rangle$$

- Unitary operators preserve inner products:

$$\langle\phi'|\psi'\rangle = \langle\phi|\hat{U}^\dagger\hat{U}|\psi\rangle = \langle\phi|\hat{I}|\psi\rangle = \langle\phi|\psi\rangle$$

- Unitary operations are reversible:

$$\hat{U}^\dagger|\psi'\rangle = \hat{U}^\dagger\hat{U}|\psi\rangle = \hat{I}|\psi\rangle = |\psi\rangle$$

  - In classical computation, AND operator is irreversible.
  - In quantum computation, AND operator is implemented reversibly using a third (ancilla) qubit.

- Question: Are there classical operations that are not allowed in QM?

Can we copy a Qubit?

- Classical information can be copied:

$$\text{bit } 0 \rightarrow 00, \qquad \text{bit } 1 \rightarrow 11$$

- Can we do the same for a qubit?

$$|\psi\rangle \rightarrow |\psi\rangle \otimes |\psi\rangle?$$

- We will have to start with two states as well.

$$|\psi\rangle \otimes |0\rangle \rightarrow |\psi\rangle \otimes |\psi\rangle?$$

- Is there a unitary operator that does this?

$$\hat{U}|\psi\rangle \otimes |0\rangle = |\psi\rangle \otimes |\psi\rangle?$$

  - Note that $\hat{U}$ would apply on the combined two qubit state.

No Cloning Theorem

- Suppose there exists a universal cloning operator $\hat{U}_{\mathrm{Cl}}$.
- Apply on two states:

$$\hat{U}_{\mathrm{Cl}}|\psi\rangle \otimes |0\rangle = |\psi\rangle \otimes |\psi\rangle, \qquad \hat{U}_{\mathrm{Cl}}|\phi\rangle \otimes |0\rangle = |\phi\rangle \otimes |\phi\rangle$$

- Take inner product:

$$\left[\langle\phi| \otimes \langle 0|\hat{U}_{\mathrm{Cl}}^{\dagger}\right]\left[\hat{U}_{\mathrm{Cl}}|\psi\rangle \otimes |0\rangle\right] = \left[\langle\phi| \otimes \langle\phi|\right]\left[|\psi\rangle \otimes |\psi\rangle\right]$$

- Using unitarity $\hat{U}_{\mathrm{Cl}}^{\dagger}\hat{U}_{\mathrm{Cl}} = \hat{I}$,

$$[\langle\phi| \otimes \langle 0|][|\psi\rangle \otimes |0\rangle] = [\langle\phi| \otimes \langle\phi|][|\psi\rangle \otimes |\psi\rangle]$$
$$\langle\phi|\psi\rangle \otimes \langle 0|0\rangle = \langle\phi|\psi\rangle \otimes \langle\phi|\psi\rangle$$
$$\langle\phi|\psi\rangle = \langle\phi|\psi\rangle^{2}$$

- This equality only holds if

$$\langle\phi|\psi\rangle = 0 \quad \text{OR} \quad \langle\phi|\psi\rangle = 1$$

- Therefore, a universal cloning operator does not exist.

The meaning of No Cloning Theorem

- Suppose that there is a unitary operator $\hat{U}_{\mathrm{Cl}}$ that can clone $|\psi\rangle$

$$\hat{U}_{\mathrm{Cl}}|\psi\rangle \otimes |0\rangle = |\psi\rangle \otimes |\psi\rangle$$

  - Then it will only clone those other states $|\phi\rangle$ which are
    - Either identical with $|\psi\rangle$: $\langle\psi|\phi\rangle = 1$
    - Or orthogonal to $|\psi\rangle$: $\langle\psi|\phi\rangle = 0$
- Example: Suppose $\hat{U}_{\mathrm{Cl}}$ clones $|0\rangle, |1\rangle$

$$\hat{U}_{\mathrm{Cl}}|0\rangle \otimes |0\rangle = |00\rangle, \qquad \hat{U}_{\mathrm{Cl}}|1\rangle \otimes |0\rangle = |11\rangle$$

  - This is allowed by the No Cloning Theorem.
  - Can we use the same operator to clone $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$?

$$\hat{U}_{\mathrm{Cl}}|\psi\rangle \otimes |0\rangle = \hat{U}_{\mathrm{Cl}}[\alpha|0\rangle + \beta|1\rangle] \otimes |0\rangle = \alpha|00\rangle + \beta|11\rangle$$

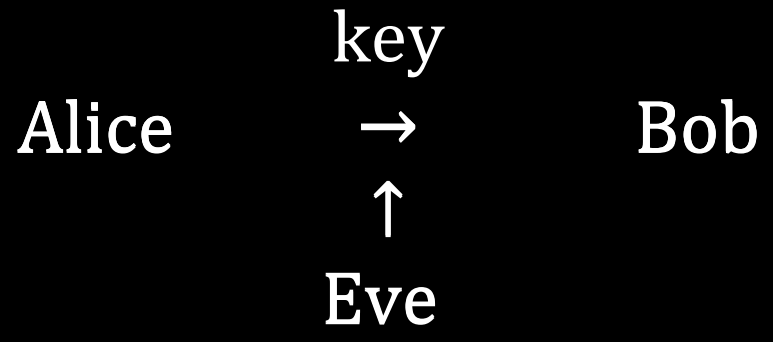
  - But cloning would require

$$\hat{U}_{\mathrm{Cl}}|\psi\rangle \otimes |0\rangle \rightarrow |\psi\rangle \otimes |\psi\rangle = [\alpha|0\rangle + \beta|1\rangle] \otimes [\alpha|0\rangle + \beta|1\rangle]$$
$$= \alpha^2|00\rangle + \alpha\beta|01\rangle + \beta\alpha|10\rangle + \beta^2|11\rangle$$

    - This would require $\alpha\beta = 0$, i.e., a state in the basis states.
    - Linearity of Quantum Mechanics prevents universal cloning.

Quantum Cryptography

- Alice wants to send Bob a secure message.
- They are concerned that Eve may eavesdrop.
- Problem: How do you secure the message?
- Solution: Send an encrypted message that unlocks with a special key.
- Problem 2: How to send the key securely?
- Eve can intercept and copy classical key.



Quantum Key Distribution (BB84 Protocol)

- Alice uses a quantum channel to send the key, meaning it is composed of multiple qubits.
- She assigns each qubit randomly in $\hat{Z}$ basis ($|0\rangle, |1\rangle$) or $\hat{X}$ basis ($|\pm\rangle$)
- Each state encodes a bit:

  bit 0: $|0\rangle$ or $|+\rangle$, bit 1: $|1\rangle$ or $|-\rangle$
- Bob measures randomly in $\hat{Z}$ or $\hat{X}$ basis.
- Alice and Bob compare bases classically.
- The matching basis results form the key.

Can Eve eavesdrop?

- Option 1: Eve copies and measures her copy

$$\hat{U}_{\mathrm{Cl}}|\psi\rangle \otimes |0\rangle = |\psi\rangle \otimes |\psi\rangle$$

  - Not allowed by No Cloning Theorem since two different basis sets are involved ($\hat{Z}, \hat{X}$).
- Option 2: Eve measures qubit and sends forward what she measured.
  - Suppose Alice sent $|+\rangle$.
  - Eve chooses right basis ($\hat{X}$) 50% of the time.
    - She measures $|+\rangle$ and knows what qubit Alice sent.
  - Eve chooses wrong basis ($\hat{Z}$) 50% of the time.
    - She measures $|0\rangle$ or $|1\rangle$.
    - She prepares that state and sends to Bob.
    - Since $|0\rangle = \frac{|+\rangle+|-\rangle}{\sqrt{2}}$ and $|1\rangle = \frac{|+\rangle-|-\rangle}{\sqrt{2}}$, Bob measures wrong 50% of the time.
  - Bob has error rate of 25%.

Secure Quantum Communication

- Without Eve: When bases match, Bob always gets correct bit.
- With Eve: Error rate of 25%.
- If error rate > 0: Eve is present.
  - Discard key and restart.
- If error rate $= 0$: key is secure.
- Quantum key distribution ensures security due to:
  - No Cloning Theorem
  - Quantum state disturbance from measurement

Can we cut and paste Qubits?

- Alice has unknown qubit:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

  - She does not know α and β.
  - Problem: She wants to send this state to Bob.
- Classical approach: Measure state and send result
  - Problem: Measurement collapses state and α, β are lost.
  - No Cloning Theorem: She cannot copy the state and send that.
- Question: Is there a cut & paste operation that can be used to send the state to Bob?
  - Answer: Yes, “quantum teleportation” of a qubit is allowed using an entangled qubit pair and two classical bits.

Quantum Teleportation: Setup

- Alice has qubit A in unknown state

$$|\psi\rangle_A = \alpha|0\rangle_A + \beta|1\rangle_A$$

- Alice and Bob share an entangled pair (qubits B and C)

$$|\phi^+\rangle_{BC} = \frac{|00\rangle_{BC} + |11\rangle_{BC}}{\sqrt{2}}$$

  - Qubit B belongs to Alice and qubit C belongs to Bob.

- Full three-qubit state:

$$|\Psi\rangle_{ABC} = |\psi\rangle_A \otimes |\phi^+\rangle_{BC}$$

$$|\Psi\rangle_{ABC} = [\alpha|0\rangle_A + \beta|1\rangle_A] \otimes \frac{|00\rangle_{BC} + |11\rangle_{BC}}{\sqrt{2}}$$

$$|\Psi\rangle_{ABC} = \frac{\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle}{\sqrt{2}}$$

- Alice will measure her two qubits A, B.

Two-Qubit Measurement

- For single qubits, measurements could be understood as being along an axis on the Bloch sphere.
  - Ex: Along z-axis, measurement gave one of the eigenstates of $\hat{Z}$

    $$|0\rangle, |1\rangle$$
  - Ex: Along x-axis, measurement gave one of the eigenstates of $\hat{X}$

    $$|+\rangle, |-\rangle$$
- For two qubits, the state space is 4-dimensional. Now the measurement basis involves 4 orthogonal states.
  - Ex: Computational Basis

    $$|00\rangle, |01\rangle, |10\rangle, |11\rangle$$
  - Ex: Bell Basis

$$|\phi^{\pm}\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \qquad |\psi^{\pm}\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}$$

Converting to Bell Basis

$$|\Psi\rangle_{ABC} = \frac{\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle}{\sqrt{2}}$$

- Suppose Alice wants to measure her qubits A, B in Bell basis

$$|\phi^{\pm}\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \qquad |\psi^{\pm}\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}$$

- Inverting the basis:

$$|00\rangle = \frac{|\phi^{+}\rangle + |\phi^{-}\rangle}{\sqrt{2}}, \qquad |11\rangle = \frac{|\phi^{+}\rangle - |\phi^{-}\rangle}{\sqrt{2}}$$

$$|01\rangle = \frac{|\psi^{+}\rangle + |\psi^{-}\rangle}{\sqrt{2}}, \qquad |10\rangle = \frac{|\psi^{+}\rangle - |\psi^{-}\rangle}{\sqrt{2}}$$

- Substituting for qubits A, B in $|\psi\rangle_{ABC}$

$$|\psi\rangle_{ABC} = \frac{1}{2}[|\phi^{+}\rangle_{AB}(\alpha|0\rangle + \beta|1\rangle)_C + |\phi^{-}\rangle_{AB}(\alpha|0\rangle - \beta|1\rangle)_C + |\psi^{+}\rangle_{AB}(\alpha|1\rangle + \beta|0\rangle)_C + |\psi^{-}\rangle_{AB}(\alpha|1\rangle - \beta|0\rangle)_C]$$

- Each term involves Alice's Bell state times Bob's superposition state involving α, β.

Quantum Teleportation

$$|\psi\rangle_{ABC} = \frac{1}{2}[|\phi^+\rangle_{AB}(\alpha|0\rangle + \beta|1\rangle)_C + |\phi^-\rangle_{AB}(\alpha|0\rangle - \beta|1\rangle)_C + |\psi^+\rangle_{AB}(\alpha|1\rangle + \beta|0\rangle)_C + |\psi^-\rangle_{AB}(\alpha|1\rangle - \beta|0\rangle)_C]$$

- Alice measures in Bell basis
  - Each outcome has ¼ probability.
  - In each case, Bob's qubit collapses to a specific state.

Alice measures $\|\phi^+\rangle$	→	Bob has $\alpha\|0\rangle + \beta\|1\rangle = \|\psi\rangle$
Alice measures $\|\phi^-\rangle$	→	Bob has $\alpha\|0\rangle - \beta\|1\rangle = \hat{Z}\|\psi\rangle$
Alice measures $\|\psi^+\rangle$	→	Bob has $\alpha\|1\rangle + \beta\|0\rangle = \hat{X}\|\psi\rangle$
Alice measures $\|\psi^-\rangle$	→	Bob has $\alpha\|1\rangle - \beta\|0\rangle = \hat{X}\hat{Z}\|\psi\rangle$

- Alice sends Bob 2 classical bits to tell him her measurement outcome.
- Bob applies corresponding inverse transformation

$$|\phi^+\rangle \to \hat{I}, \qquad |\phi^-\rangle \to \hat{Z}, \qquad |\psi^+\rangle \to \hat{X}, \qquad |\psi^-\rangle \to \hat{Z}\hat{X}$$

- Bob recovers $|\psi\rangle_C = \alpha|0\rangle_C + \beta|1\rangle_C$!

Quantum Teleportation: Conclusion

- No physical qubit is sent!
  - Quantum information is sent using entanglement.
  - The state is ‘teleported’ from one qubit to the other in a different location.
- There is no violation of No Cloning Theorem.
  - Alice’s qubit collapses after she makes her measurement.
- There is no faster than light communication.
  - Bob cannot get $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ until he gets the classical bits from Alice telling him which operations to perform.
- Entanglement is used as a resource for communication.
  - But it is consumed by the end.
  - New Bell pair needed for each qubit teleportation.

Summary

- No Cloning Theorem forbids the existence of a universal cloning operator of the form $\hat{U}_{\mathrm{Cl}}|\psi\rangle \otimes |0\rangle = |\psi\rangle \otimes |\psi\rangle$.
- BB84 Protocol allows secure quantum key distribution by detecting eavesdropping.
- Quantum Teleportation allows sending qubits using quantum entanglement.
- Qubits cannot be copy and pasted.
- Quantum information can be securely communicated using classical bits.
- 1 Bell state + 2 bits sent → 1 qubit transmitted

Quantum resource | Classical Channel | Quantum Information